Passive ISAC for Urban Transit using Self-Supervised World Models

Opportunistic Wi-Fi Sensing for Passenger Monitoring

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The inMotion Project

The **inMotion** project aims to bring real-time intelligence to public transport operations.
One of its core goals, understand **how passengers move** through the network: who boards, who alights, and where.
A previous paper answered that with hand-crafted RSSI features and supervised classifiers. Today the question is different: can a **world model learn the dynamics** itself?



Smart bus concept

The Passenger Counting Gap

Without granular passenger data:

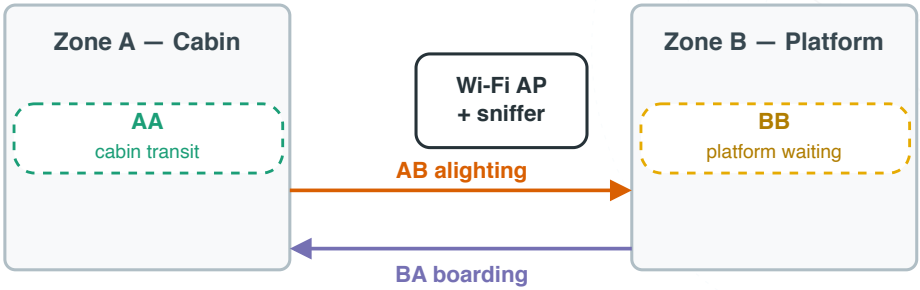
- Routes designed on rough estimates, not actual demand
- Buses run near-empty on some segments, overcrowded on others
- Operators cannot justify service adjustments to funders
- Fleet sizing based on guesswork, not occupancy

The potential of data-driven optimization:

Metric	Improvement
Route costs	↓ **30–40%**
Fuel consumption	↓ **10–20%**
Routes needed	↓ **29–42%**

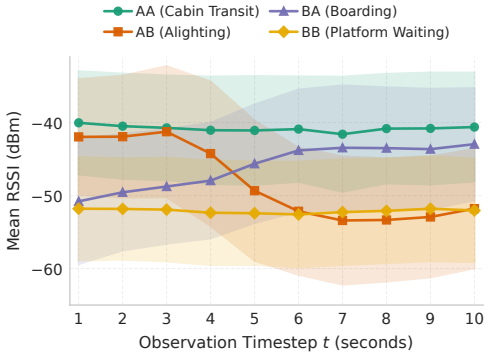
Sources: UPS ORION (30M liters fuel/year saved); industry route optimization benchmarks

Four Movements to Classify



Four states: **AA** cabin transit · **AB** alighting · **BA** boarding · **BB** platform waiting.
One observation: 10 s at 1 Hz, so 10 RSSI values per window.

Wi-Fi RSSI Is a Movement Signature



Mean RSSI per class on the 3,511-sample collection ($\pm 1\sigma$)

The trend carries the class:

- **Boarding (BA):** signal rises from -51 to -43 dBm as the device approaches the AP
- **Alighting (AB):** falls from -42 to -52 dBm as it walks away
- **Cabin transit (AA):** stable at ≈ -40 dBm
- **Platform waiting (BB):** stable at ≈ -52 dBm

The $\pm 1\sigma$ bands overlap heavily. Classes differ in **shape**, not in level.

How a World Model Learns

1 Watch



Ten seconds of radio
from one access point.
No camera, no label.

2 Guess



Hide the second half.
Guess it — as an idea,
not as a waveform.

3 Understand

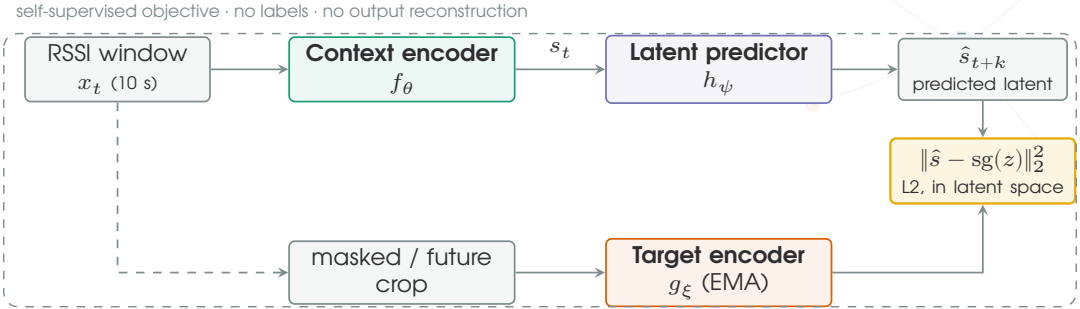


What stays is motion:
toward the door, or away.

It never sees the answer. That alone beats every supervised model here — and it keeps getting better as we add data.

The world here is the radio field around the door; the physics is a passenger walking.

Predicting in Latent Space



No decode back to the signal. The target encoder is a slow EMA copy of the context encoder and its output is stop-gradiented, so the two branches cannot collapse onto each other; the loss is an L2 distance between two *latent* vectors.

Data Collection Setup



Controlled indoor environment, IT Aveiro

Physical setup:

- Zone A: closed room (cabin), AP at the doorway
- Zone B: corridor (platform), wall attenuation between zones

Collection:

- 10 s windows at 1 Hz $\rightarrow$ 10 RSSI values
- **7 smartphone models**, multiple vendors, multiple Android releases
- Scripted movements with ground-truth session labels
- Passive sniffing: no association, no identification

Dataset Composition and Splits

Condition	Samples	Description
Single-device	160	pure signals, 40 per class
Multi-device	3,351	concurrent interference, ≈ 840 per class
Total	**3,511**	primary benchmark

Strict superset. The earlier 2,251-sample collection is fully contained: all eight earlier devices are retained, and five more are added.

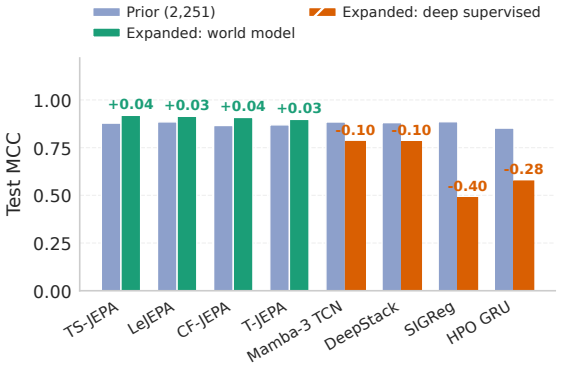
Protocol. 80/20 class-stratified split, with 12.5% of training held out for validation. Gaussian noise ($\sigma = 0.05$) and intra-class Mixup ($\alpha = 0.4$) give **35,839** training windows; the test split is evaluated once.

The Four World Models

	TS-JEPA	LeJEPA	CF-JEPA	T-JEPA
Input channels	18 (rich)	4 (base)	18 (rich)	18 (rich)
Tokenizer	1D conv patch	causal temporal	1D conv patch	per-feature + REG
Patch / token size	2 timesteps	1 timestep	2 timesteps	1 timestep
Embedding dim / heads	256 / 8	128 / 4	256 / 8	128 / 4
Encoder layers	4	4	3	4
Predictive objective	masked patch L2	next-step L2	multi-horizon L2	masked token L2
Anti-collapse	EMA target	SIGReg	EMA target	EMA + REG
Parameters	2.153 M	**0.566 M**	1.232 M	0.633 M

Baselines: Mamba-3 state spaces, four front-ends (0.84–1.70 M) · **SIGReg**: directly supervised CNN with the SIGReg regulariser, *no pretraining* (0.697 M) · DeepStack (43.6 M) · Optuna-tuned GRU.
Interpretability: SHAP attributions over the 18 channels, plus architecture-controlled ablation.

MCC Across Model Families



Prior (2,251) vs expanded (3,511); the number above each pair is Δ MCC

Expanded collection, test MCC:

- **TS-JEPA: 0.9194**
- LeJEPA: 0.9143
- CF-JEPA: 0.9083
- T-JEPA: 0.8982
- Mamba-3 TCN: 0.7816
- DeepStack: 0.7811

Every world model clears the previous **0.756** Gaussian Process baseline by a wide margin.

Stability Under Collection Expansion

Model	Prior	Expanded	Δ
TS-JEPA	0.8785	**0.9194**	$+0.041$
LeJEPA	0.8850	0.9143	$+0.029$
CF-JEPA	0.8661	0.9083	$+0.042$
T-JEPA	0.8694	0.8982	$+0.029$
Mamba-3 TCN	0.8844	0.7816	-0.10
DeepStack	0.8810	0.7811	-0.10
SIGReg	0.8858	0.4872	-0.40
HPO GRU	0.8524	0.5746	-0.28

The contrast is the finding:

- Self-supervised world models: $+0.029$ **to** $+0.042$
- Every deep supervised model: **degrades**, up to -0.40

SIGReg has no pretraining phase — it is fitted directly on labels. A richer interference regime falls outside what it learned.

Same device pool, so this measures **stability under collection expansion**, not transfer to a new environment.

Stacking and Calibration

Ensemble	n	Params	Prior	Expanded	Noise / Pure
21-member full stack	21	74.8 M	0.8873	**0.9232**	**0.9563** / **0.9596**
16-member subset	16	73.2 M	0.8873	**0.9232**	**0.9563** / 0.9179
World-only stack	7	6.68 M	0.8872	0.9065	0.9450 / **0.9596**
Best-world subset	3	3.42 M	0.8784	0.9160	0.9341 / 0.8773
Three-member Pareto	3	**1.26 M**	**0.8904**	0.5753	0.9396 / 0.9179

- Every member is **temperature-calibrated** on validation NLL, then fused by an L2 multinomial logistic-regression meta-learner.
- The 16-member non-redundant subset matches the full 21-member stack exactly: the weak recurrent backbones add nothing.
- Ensembles are pruned by a greedy **Pareto search**: MCC against parameter count.

Accuracy versus Model Size

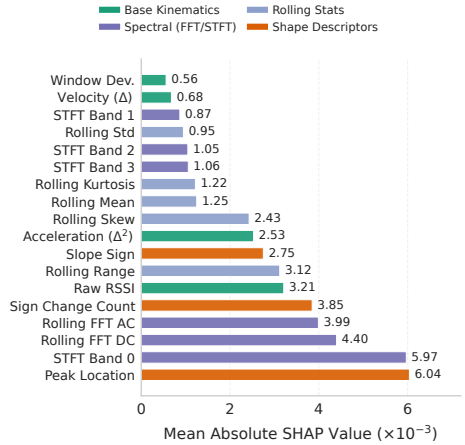
74.8 M
21-member stack
0.9232 MCC
peak accuracy

3.42 M
3-member subset
0.9160 MCC
-0.007 for $22\times$ fewer

0.566 M
LeJEPA alone
0.9143 MCC
on 4 channels

Two deployment paths. Rich 18-channel tokenisation needs the full hierarchy; a compact model designed for four kinematic channels nearly matches it, which is what makes an access-point implementation plausible.
The catch: the 1.26 M frontier was picked for size on the *earlier* collection (logreg, LeJEPA, SIGReg) and does not carry over — 0.8904 there, 0.5753 here. Keeping TS-JEPA in the mix instead (3.42 M) holds at 0.9160.

Channel-Level Attribution



Mean absolute SHAP value per channel, TS-JEPA, 3,511-sample test split

What drives the prediction:

- **Peak location** (6.04×10^{-3}): *when* the door transition happens
- **STFT band 0** (5.97×10^{-3}): coarse Doppler of physical motion
- **FFT DC / AC** (4.40 / 3.99×10^{-3}): energy shift between zones
- **Everything else** (1.22 kurtosis, 0.95 std, high STFT): multipath, not motion

What a bar means. A Shapley value is a channel's average marginal contribution over every coalition of the others; bars are mean $|\phi_j|$ on the test split (Lundberg & Lee, NeurIPS 2017).

Next slide: the same test, controlled architectures.

Feature-Channel Ablation

Feature configuration	C	TS-JEPA	CF-JEPA	T-JEPA
Full 18-channel hierarchy	18	**0.9194**	**0.9083**	**0.8982**
Top-12 SHAP channels	12	0.5815	0.8098	0.5856
Base kinematics (raw, Δ, Δ^2, dev)	4	0.4486	0.4366	0.4787
Rolling statistics (moments, range)	5	0.3447	0.4351	0.4944
Spectral domain (FFT, STFT 0–3)	6	0.2981	0.1869	0.2283
Raw RSSI only (single channel)	1	0.2431	0.3305	0.3395

Restricting a fixed architecture to any single functional domain destroys it: TS-JEPA falls from 0.9194 to 0.4486 on the four base channels, and to 0.2431 on raw RSSI. Even the twelve most important channels, selected by SHAP, are not enough: the synergy is not additive.

Privacy Through Edge Inference

The threat is real:

- Passive sniffing needs no cooperation and no visible trace
- Radio sequences survive MAC randomisation
- Commuter regularity links a passenger across days
- Profiling defeats naive anonymisation and DP (Silva et al.)

What the architecture buys:

- 0.9143 MCC at **0.566 M** parameters: small enough for AP firmware
- Design target: infer *inside* the AP (deployment is future work)
- Raw RSSI and latents live in RAM and are discarded
- Only aggregated counts leave: Δboarding, Δalighting

The accuracy that makes the sensing useful is the same property that makes edge confinement plausible.

What These Results Mean

What we have demonstrated:

- **Four movement classes** from 10 RSSI values, no dedicated hardware
- **0.9194** single model · **0.9232** calibrated ensemble
- **0.9563** under multi-device interference
- **Stability**: world models gain as data grows, supervised models break
- **0.9160** at 3.42 M parameters, edge-feasible

What we must be careful about:

- Two **controlled indoor zones**, not a vehicle in service
- Seven phone models here, thirteen across campaigns; no demographics recorded
- Labels from the **session protocol**, not independent annotation
- The collections **share their device pool**: gains measure stability, not transfer
- 160 single-device traces are too few for a separate held-out pure evaluation

Conclusions and Future Work

This work:

- **Self-supervised world models** for passive ISAC in transit
- Four JEPA variants, 18-channel hierarchy, calibrated ensembles
- **0.9194** single · **0.9232** ensemble · **0.9160** at 3.42 M
- **Stability under collection expansion**, where supervised models collapse

Models and code:
github.com/ATNoG/inMotion

Next steps:

- **Cooperative multi-AP fusion** with spatial diversity
- **Online continual adaptation** to seasonal radio drift
- **Elastic privacy**: throttle resolution by motion state
- **Operational validation** on a vehicle in service
- Close the gap between single models (0.9194) and the 0.9563 ensemble result on the multi-device subset

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Questions?

Code and models: github.com/ATNoG/inMotion · Paper: FNWF 2026